

Name:	
Class:	
Date:	

Honor pledge <i>“On my honor, I have neither given nor received unauthorized aid.”</i> Student Signature: _____
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Subject:	IB1 Computer Science SL + HL
Exam:	G11 T1 A1 (B2.1)
Exam Duration:	60 Minutes
Teacher:	Mr Ellis

Instructions to Students:
Answer all questions. Write your answers in the spaces provided. The number of marks is shown in brackets []. No calculators permitted. Exam completed in 30 minutes Next assessment move to 45 minutes and 60 marks, Give more difficult command term questions Consider whether to include a revision guide. Total marks available for this exam = 40

IB Secondary School Grade Conversion Matrix		
IB Score	ISR Percentage Grade	Descriptive
7	90-100	Outstanding
6	85-89	Excellent
5	80-84	Good
4	70-79	Satisfactory
3	60-69	Needs Improvement
2	45-59	Unsatisfactory
1	0-44	Unsatisfactory

Marks:
ISR % Grade:
IB Score:

Grade 11 Assessment 1 – B2.1 Programming Fundamentals

Aligned with IB Computer Science (First Assessment 2027)

Section A: Variables & Data Types

(8 marks)

1. State the data type that would be most appropriate for each of the following values in Python:

[4]

Value	Data Type
25	
3.14	
"Hello"	
True	

2. A variable is declared inside a function in Python.

(a) State the **scope** of this variable.

[1]

(b) Explain why this variable **cannot** be accessed from outside the function. [2]

3. A programmer wants to modify a **global** variable inside a function.

State the keyword that must be used to achieve this. [1]

Section B: String Manipulation

(10 marks)

4. A string `word = "Computer"` has been declared in Python.

State the output of each of the following expressions: [3]

Expression	Output
<code>word[0:3]</code>	
<code>word[3:]</code>	
<code>word[-3:]</code>	

5. A programmer writes the following code:

```
greeting = "Hello"
name = "Bill"
full = greeting + ", " + name + "!"
print(full)
```

State the output of this code. [2]

6. Study the following code:

```
message = "Hello, World!"
print(message.lower())
print(message.replace("World", "IB"))
```

State the output of each line. [2]

Line 1:

Line 2:

7. Explain why strings in Python are described as **immutable**.

[3]

Section C: Exception Handling

(10 marks)

8. State the purpose of each of the following keywords used in exception handling: [3]

Keyword	Purpose
<code>try</code>	
<code>except</code>	
<code>finally</code>	

9. Study the following code:

```
try:
    number = int(input("Enter a number: "))
    result = 10 / number
    print(result)

except ValueError:
    print("Please enter a valid integer.")

except ZeroDivisionError:
    print("You cannot divide by zero.")
```

(a) State the output if the user enters `"abc"`.

[1]

(b) State the output if the user enters `"0"`. [1]

(c) Explain why `except ZeroDivisionError` would **not** work if it were placed **after** `except Exception`. [2]

10. A programmer is writing a program that reads data from a file. The file may not exist.

Explain how the `try` and `except` blocks could be used to handle this situation. [3]

Section D: Debugging Techniques (12 marks)

11. Distinguish between the following types of errors, giving one example of each: [4]

(a) Syntax error

(b) Logical error

12. A programmer has written the following code:

```
total = 0
for i in range(3):
    total += i
print(total)
```

(a) Complete the trace table below for this code. [3]

Iteration	i	total
1		
2		
3		

(b) State the final output of the program. [1]

13. A programmer is debugging a program that crashes with a `TypeError` when processing a list of prices: `[10, 20, "30", 40]`.

Describe how a **print statement** could help identify the problem. [2]

14. Explain what a **breakpoint** is and how it helps in debugging. [2]

END OF EXAM

Total: 40 marks